Venkata Lakshmi Parimala Pasupuleti

Principal GenAI & Agentic Systems Engineer · Python · LangGraph · LLM Orchestration · Microservices · RAG · On-Prem + Cloud LLMs · Financial Data at Scale · Production AI

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PROFESSIONAL SUMMARY

GenAI and agentic systems engineer with 9+ years delivering production AI, microservices architecture, and financial data pipelines at scale. Built **AgentProbe**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, multi-provider LLM routing across **cloud (Claude, OpenAI, Groq) and on-premises (Ollama)**, 5-metric eval engine, CI/CD regression gates, and React observability dashboard. Built **Leasel**: agentic RAG pipeline achieving 93.6% precision over **unstructured documents** with a deterministic rules engine for auditable outputs. At Entrata: **production AI pipeline on 1.6M financial records** (Python, PyTorch, OpenAI API); **microservices architecture redesign** (50% scalability, 30% deployment reduction); **200+ SQL optimizations**; 100K+ MAU, 99.5% uptime. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Present

Graduate Teaching Assistant & Research Associate — Python · Machine Learning Pipelines · Data Analytics · Anomaly Detection

- Mentored **600+ students** across Python programming, database design, SQL, and data analysis (MIS 303); built multi-source ETL pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 machine learning research pipelines; designed analytics dashboards driving **40% improvement** in decision comprehension.

Entrata — Real Estate PropTech SaaS
Sep 2018 – Aug 2024

Senior Software Engineer — Production AI · Microservices · Financial Data at Scale · Python · REST APIs · AWS

- Shipped **Entrata's first production AI pipeline on 1.6M financial records** (Python, PyTorch, OpenAI API) — custom neural network architecture for transformation and validation of **structured and unstructured financial datasets**; automated data processing workflows replacing manual Excel operations with output validation, quality gates, and compliance controls.
- Led **microservices architecture redesign with Docker** — 50% improvement in system scalability, 30% reduction in deployment time; designed multi-tier service-based architecture across 20+ production modules; **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, partitioning, caching, triggers, audit tables; led database architecture decisions for complex production schemas.
- Built **REST API integrations and inter-service communication** (RabbitMQ, AWS S3/EC2/RDS/Glue) meeting **scalability and availability requirements** across 100K+ MAU, 99.5% uptime, 120+ Agile sprints; achieved **100% PHI compliance** (Libsodium encryption); eliminated 80% manual workload via Python/Bash automation; **300+ code reviews, 5+ engineers mentored**; stakeholder partnership with data architects, QA, and product teams across organizational boundaries.

Entrata India
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · REST APIs · AWS S3

- Built full-stack modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts for 1.6M financial records; developed data normalization pipelines, REST API integrations, and AWS S3 storage; participated in design reviews, sprint planning, and code quality processes.

PROJECTS

AgentProbe: Scaling Agentic Solutions - Cloud + On-Prem LLM Routing & Eval
AGENTIC SYSTEMS · SCALABILITY

Python · LangGraph · LangChain · Claude API · OpenAI · Groq · Ollama (on-prem) · FastAPI · ChromaDB · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- **3-agent LangGraph runtime** with tool calling, inter-agent state handoffs, and **multi-provider LLM routing across cloud (Claude, OpenAI, Groq) and on-premises (Ollama)** — designed for environments where sensitive data cannot leave the network; **5-metric LLM-as-judge eval engine** with CI/CD regression gates on every commit; React **observability dashboard** tracking per-run metrics, latency, and retrieval drift automatically.

Leasel: Agentic RAG over Unstructured Documents with Deterministic Output Validation
GENAI · LLM EVALUATION

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 **unstructured** residential lease PDFs — agentic RAG pipeline with vector retrieval and LLM reasoning; deterministic rules engine validates all outputs before writing to database; pass@k eval harness (pass@1 81.7%, pass@3 88.3%); CI/CD regression gate; scalable to any document corpus.

AI on FHIR: NLP-Driven Medical Records Query System
ENTERPRISE AI

Python · Flask · React · LLMs · BERT · Embeddings · FHIR APIs · Docker · HIPAA-compliant RBAC — github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP platform translating natural-language queries to FHIR APIs using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records; reduced manual retrieval time ~60%; demonstrates LLM deployment over sensitive structured enterprise data.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

ML systems, AI pipelines, data engineering, statistical modeling, anomaly detection, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

GenAI & Agentic Systems: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Scaling Agentic Solutions, Tool Calling, LLM Orchestration, RAG Pipelines, Vector Databases, ChromaDB, Semantic Search, Generative AI Implementation, LLM Evaluation, LLM-as-Judge, Eval Frameworks, Deterministic Output Validation, Agent Observability, Prompt Engineering, Claude API, OpenAI API, Groq, Ollama (on-premises LLM), FastAPI

Engineering & Architecture: Python (expert, OOP), SQL (expert — 200+ production optimizations, execution plan tuning, indexing, partitioning, audit tables), REST API Development, Microservices Architecture, Multi-Tier Platforms, Service-Based Architecture, Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, PostgreSQL, MySQL, Libsodium, PHP, JavaScript, TypeScript, Linux, Git

ML & Data: PyTorch, scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Feature Engineering, NLP, BERT, ETL Pipelines, Data Modeling, Structured + Unstructured Dataset Processing, Tableau, Power BI, Statistical Modeling, Agile/Scrum